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Relationship between trunk and pelvic rotation angles during throwing and the Kerlan−Jobe Orthopedic Clinic score in American football quarterbacks

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Abstract

Background: Quarterbacks (QBs) in American football frequently perform repetitive overhead throwing, which can lead to shoulder and elbow injuries. Effective control of trunk and pelvic movements is considered crucial for minimizing shoulder and elbow joint stress during throwing. The Kerlan–Jobe Orthopedic Clinic (KJOC) score is a self-perceived index of shoulder and elbow function and level of competition in overhead athletes. However, the relationship between trunk and pelvic movements during throwing and the KJOC score among QBs remains unclear. This study aimed to examine the association between trunk and pelvic motion during the QB throwing sequence and the KJOC score.

Methods: A total of 11 healthy QBs were enrolled in this study. Each participant completed five throws at distances of 20 and 30 yards. Throwing mechanics were analyzed using the PitchAI™ application, focusing on trunk and pelvis rotation (toward the nonthrowing and throwing sides) and hip–shoulder separation angles. These angles were measured at three key phases: stride foot contact (SFC) during the early cocking phase, maximum shoulder external rotation (MER) during the late cocking phase, and ball release (BR). Participants also completed the KJOC questionnaire, with higher scores indicating better self-perceived index of shoulder and elbow function or level of competition . Correlations between throwing mechanics and KJOC each item and total scores were assessed using Spearman's rank correlation coefficient.

Results: Trunk rotation at MER during 20-yard throws and hip–shoulder separation at SFC during 30-yard throws were positively correlated with KJOC Item 3 ($r_s = 0.709$; $p = 0.015$ and $r_s = 0.662$; $p = 0.026$, respectively). Pelvic rotation at BR during 30-yard throws showed a positive correlation with KJOC Item 4 ($r_s = 0.760$; $p = 0.007$).

Conclusion: This study suggests a relationship between trunk and pelvic rotational movements

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during a quarterback's throw and certain aspects of subjective shoulder and elbow function.

Level of evidence: Basic Science Study; Validation of Outcome Instrument

Keywords: Throwing motion analysis, trunk movement, pelvis movement, American football, quarterbacks, Kerlan–Jobe Orthopedic Clinic score

American football is a widely popular sport in the United States and has gained significant popularity in Japan. Among the key positions in the sport is the quarterback (QB), who initiates offensive plays by throwing a football, typically weighing between 397 and 425 g, to a teammate. However, repeated throwing exposes QBs to a risk of overuse injuries. Radel et al[13] reported that approximately 15% of QBs under the age of 18 sustained shoulder impingement syndrome, a common shoulder injury. In the same study, they also identified medial collateral ligament injuries and osteochondritis dissecans of the humerus as prevalent elbow injuries among QBs.[13] Shoulder and elbow injuries requiring surgical intervention typically demand a recovery period of approximately 7–8 months before athletes can return to play.[4,11] Given these findings, it is essential to implement measures aimed at preventing throwing-related shoulder and elbow injuries and maintaining optimal joint function to preserve the athletic performance and career longevity of QBs.

The loads placed on quarterbacks' shoulders and elbows during throwing have been investigated by several researchers.[8,14] Fleisig et al[8] reported that during the cocking phase, quarterbacks generate a shoulder anterior force of 350 N and an elbow varus torque of 54 Nm, while Rash et al observed a maximum shoulder anterior force of 250 N. These findings indicate that throwing imposes substantial stress on the shoulder and elbow joints. Quarterback throwing is a whole-body motion in which energy produced by the lower limbs and trunk is transmitted to the upper limbs. In overhead sports, it is generally recognized that inadequate kinetic chain function in proximal

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segments, such as the trunk and pelvis, increases the mechanical load on distal segments, including the shoulder and elbow.[3,9] Furthermore, Labbe et al[10] further demonstrated, using inertial measurement units, that trunk and pelvic rotation occurs from stride foot contact to ball release, and that efficient energy transfer to the upper limbs requires appropriate rotation of these segments. Collectively, these reports suggest that controlling trunk and pelvic rotation during throwing may help reduce shoulder and elbow loading and support joint function.

The Kerlan–Jobe Orthopedic Clinic (KJOC) score is used to evaluate a self-perceived index of shoulder and elbow function and level of competition in overhead athletes.[2] The KJOC score is a questionnaire-based assessment that evaluates not only shoulder and elbow symptoms but also competitive level, and has demonstrated greater sensitivity than the Disabilities of the Arm, Shoulder and Hand (DASH) questionnaire and its sports/performing arts module for assessing function in overhead athletes.[2] Trunk stability has been identified as a contributing factor influencing the KJOC score.[5] Chassé et al[5] used the Upper Quarter Y-Balance Test as an indicator of trunk stability and reported a positive correlation between test performance and KJOC score in baseball pitchers. While this study examined the relationship between physical function and KJOC scores, it did not evaluate the association between KJOC scores and throwing mechanics. Demonstrating a relationship between trunk and pelvic rotational movements during a quarterback's throw and the KJOC score may provide a useful reference point when coaching throwing mechanics in quarterbacks with subjectively reduced shoulder and elbow function.

Therefore, this study aimed to examine the relationship between trunk and pelvic rotational movements during the throwing motion in QBs and their KJOC scores. The hypothesis was that trunk and pelvic rotational movements during throwing are correlated with KJOC scores.

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Materials and Methods

Participants

Participants were quarterbacks from a university American football team or a club team affiliated with the Japan American Football League Xcellence. Recruitment was conducted through the university team associated with the authors' institution and a club team previously attended by one of the co-authors (S.A.). Inclusion criteria required players to be capable of throwing at maximum effort and to fully participate in quarterback throwing training. Exclusion criteria included a history of shoulder or elbow surgery, as well as current pain in the lower back or lower extremities. During recruitment and testing, athletes were verbally screened for pain or symptoms in the limbs and lower back. The study protocol was approved by the Research Ethics Committee (study number: 2023-301-A). Written informed consents were obtained from all participants before data collection.

Throwing motion analysis

Before the throwing test began, participants were given unlimited time to perform warm-up and stretching exercises. The throwing test was conducted on a football field. Each participant performed five maximum-effort throws with an official football (400–425 g) toward a 1.1 × 1.1 m target at distances of 20 and 30 yards, which reflect common passing ranges for quarterbacks. Preliminary checks confirmed that some players were unable to throw accurately beyond 40 yards, whereas throws at 10 yards could be executed without trunk or pelvic rotation. Accordingly, two distance conditions—20 yards and 30 yards—were selected for this study.

For motion analysis, an iPad (Apple Inc., Cupertino, CA, USA) was positioned laterally so that the participant's entire body could be seen from the side. The device was mounted on a tripod to prevent panning. Recording settings were 240 Hz frequency, 1080 p resolution, and landscape orientation. After capturing the throwing motion, the PitchAI (3Motion AI Inc, Oakville, Canada)

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was used to analyze trunk rotation, pelvic rotation, and hip–shoulder separation angles at three key events: SFC during the early cocking phase, maximum shoulder external rotation (MER) during the late cocking phase, and ball release (BR) (Figure 1).[6,8] Trunk and pelvic angles were calculated as absolute values relative to a coordinate system with the flat ground serving as the reference plane.[6] Trunk rotation was defined as 0 degrees when the line connecting both shoulders was parallel to the line from the pivot foot to the target. Positive values indicated trunk rotation toward the nonthrowing side. Similarly, pelvic rotation was defined as the angle between the line connecting both hip joints and the line from the pivot foot to the target, with 0 degrees indicating parallel alignment and positive valuesindicating pelvic rotation toward the nonthrowing side. The hip–shoulder separation angle was calculated as the difference between pelvic and trunk rotation angles, with positive values indicating trunk rotation toward the throwing side relative to the pelvis. PitchAI uses neural networks to convert two-dimensional images into three-dimensional motion data using a markerless motion capture system. Furthermore, the application automatically calculates joint angles from the uploaded video. The validity of trunk and pelvic rotation angles measured by PitchAI has previously been examined in baseball pitching, and these measures have been reported to be sufficiently valid.[6] Therefore, in this study, we used PitchAI due to its ease of use for analyzing pitching motion in outdoor settings.

Kerlan–Jobe Orthopedic Clinic score and demographic data of the participants

All participants completed a paper-based KJOC questionnaire during the test session. This instrument consists of 10 items assessing shoulder and elbow health, function, and competition level (Table I).[2] Participants were instructed to mark an "X" on a 10-cm line corresponding to each question. The position of the mark indicated their current level of competition, functioning, or both,

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with marks placed closer to the right end indicating higher ratings. Scores for each item and the total KJOC score were then calculated accordingly.

In addition to the KJOC score survey, participants' age, body mass, height, and experience were also surveyed. These demographic data were self-reported on paper.

Statistical analysis

Statistical analyses were performed using JMP Pro software, version 16 (SAS Institute Inc., Cary, North Carolina, USA). Normality of the KJOC score, trunk rotation angle, pelvic rotation angle, and hip–shoulder separation angle was assessed using the Shapiro–Wilk test. Relationships between the KJOC score and the three kinematic variables during the throwing motion were examined using Pearson's correlation coefficient for normally distributed data and Spearman's rank correlation coefficient for non-normally distributed data. Statistical significance was set at p of <0.05. Interpretation of Spearman's correlation coefficients (rs) followed these thresholds: <0.10, no correlation; 0.10–0.39, weak; 0.40–0.69, moderate; 0.70–0.89, strong; and ≥ 0.90, very strong.[16] For Spearman's rank correlation, a sample size of 84 participants was calculated to detect an effect size of 0.30 with a significance level of 0.05 and a statistical power of 0.8.

Results

In this study, 11 quarterbacks volunteered and all completed the measurements. Participant demographics are shown in Table II. Normality was not observed for individual and total KJOC scores, the pelvic rotation angle at BR during the 20-yard throw, the pelvic rotation angle at SFC during the 30-yard throw, and the hip–shoulder separation angle at MER during the 30-yard throw.

Data on trunk rotation, pelvic rotation, and hip–shoulder separation angles during throwing and KJOC scores are shown in Tables III and IV, respectively. Their relationships are summarized in Tables V and VI. For the 20-yard throw, a positive strong correlation was observed between KJOC

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Item 3 and trunk rotation angle at MER (r_s = 0.709; p = 0.015). KJOC Item 7 was positively moderate correlated with pelvic rotation angle at BR, and KJOC Item 10 was positively moderate correlated with pelvic rotation angle at MER (r_s = 0.658; p = 0.028 and r_s = 0.616; p = 0.044, respectively).

For the 30-yard throw, KJOC Item 3 showed a positive moderate correlation with the hip–shoulder separation angle at SFC (r_s = 0.662; p = 0.026). KJOC Item 4 was positively strong correlated with pelvic rotation angle and hip–shoulder separation angle at BR (r_s = 0.760; p = 0.007 and r_s = 0.779; p = 0.005, respectively). KJOC Item 5 was positively strong correlated with hip–shoulder separation angle at BR (r_s = 0.752; p = 0.008). KJOC Item 7 was positively moderate or strong correlated with trunk rotation, pelvic rotation, and hip–shoulder separation angles at BR (r_s = 0.687; p = 0.020, r_s = 0.839; p = 0.001, and r_s = 0.734; p = 0.010, respectively). KJOC Item 10 was positively moderate correlated with the hip-shoulder separation angle at SFC (r_s = 686; p = 0.020). No significant correlations were found with other items.

Discussion

This study aimed to examine the relationship between trunk and pelvic rotational movements during quarterback throwing and KJOC scores. We hypothesized that these kinematic variables would correlate with KJOC scores. The main finding was a positive correlation between trunk and pelvic rotation during throwing and KJOC Items 3 and 4, suggesting that quarterbacks' subjective shoulder and elbow function is related to their trunk and pelvic mechanics.

In regards to the influence of trunk and pelvic rotation during SFC, Douoguih et al[7] reported that early trunk rotation in the initial cocking phase may be associated with shoulder and elbow injuries requiring surgical intervention. Similarly, Lin et al[12] classified pitchers into early and non-early trunk rotation groups based on the pelvic–trunk rotation angle at SFC, defining early

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rotation as a rotation difference of <25°.[12] The early rotation group demonstrated increased horizontal abduction and reduced abduction of the shoulder joint after MER.[12] These movements are known to increase stress on the glenohumeral joint, and previous research suggests that early trunk rotation during SFC may affect shoulder joint mechanics.[1,17]

In the present study, the hip–shoulder separation angle was used to represent trunk rotation relative to the pelvis, with smaller angles reflecting greater trunk rotation toward the non-throwing side. Thus, this measure at SFC served as an indicator of early trunk rotation. A significant correlation was found between the hip–shoulder separation angle at SFC and Item 3 of the KJOC score, suggesting that the timing of trunk rotation may be associated with quarterbacks' subjective perception of shoulder and elbow function.

Reports examining the effect of trunk and pelvic rotation after MER on shoulder and elbow function are limited. Throwing is a kinetic chain movement that transfers energy from the lower limbs through the pelvis and trunk to the upper limbs, and it is generally believed that restricted trunk and pelvic rotation can increase mechanical stress on distal segments.[3,9] Howenstein et al[9], using three-dimensional motion analysis, reported that limited trunk and pelvic rotation following the late cocking phase may be associated with increased shoulder stress. Similarly, Sakata et al[15] identified reduced pelvic rotation during pitching as a risk factor for shoulder and elbow injuries, emphasizing the importance of rotational mechanics in throwing.

In our study, significant correlations were observed between trunk and pelvic rotation angles after MER and KJOC Items 3 and 4. These findings suggest that rotational movement after MER may be associated with quarterbacks' subjective perceptions of shoulder and elbow function. However, because this analysis was correlational, causality and kinetic mechanisms cannot be established. Further longitudinal and biomechanical studies are required to clarify the role of

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trunk and pelvic rotation in subjective shoulder and elbow function.

This study has several limitations. This study has several limitations. First, the sample size was small, with quarterbacks recruited from only two teams due to existing affiliations of the authors and co-authors. The limited number of participants may have reduced statistical power. Nonetheless, Spearman's rank correlation coefficients indicated moderate or stronger associations for some variables, suggesting that meaningful correlations between throwing mechanics and KJOC scores exist despite the small cohort. Future studies with larger sample sizes are necessary to confirm these findings. Second, the validity of trunk and pelvic angle measurements during quarterback throwing remains uncertain. Dobos et al[6] compared trunk and pelvic angles measured during baseball pitching using a marker-based motion capture system and PitchAI and found no significant differences, supporting the validity of PitchAI measurements.[6] However, its validity and reliability for quarterback throwing have not yet been examined. Future research should investigate the accuracy of this tool in football-specific contexts. Third, the distribution of KJOC scores may have introduced a ceiling effect. As presented in Table IV, most items were rated very highly, reflecting excellent shoulder and elbow function among participants. The limited variability in scores may have reduced sensitivity to subtle differences or changes in function. More sensitive assessment tools should be considered in future studies to better capture nuances in shoulder and elbow function among quarterbacks.

Conclusions

In this study, we examined the relationship between the throwing motion of QBs and their KJOC scores. This study suggests a relationship between trunk and pelvic rotational movements during quarterback throwing and certain aspects of subjective shoulder and elbow function.

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Figure 1. Phases of throwing analyzed

(A) Stride foot contact during the early cocking phase. (B) Maximum external rotation of the shoulder during the late cocking phase. (C) Ball release

Table I. Kerlan–Jobe Orthopedic Clinic (KJOC) score

Table II. Demographic data of the participants

SD, standard deviation

Table III. Trunk, pelvic, and hip–shoulder separation angles during throwing at 20 and 30 yards

Data are presented as the median and quartile deviation. Values in brackets represent the quartile deviation.

Table IV. Total KJOC score and individual item score

Data are presented as the median and quartile deviation. Values in brackets represent the quartile deviation.

Table V. Correlation between trunk, pelvic, and hip–shoulder separation angles during 20-yard throws and KJOC scores

SFC, stride foot contact during the early cocking phase; MER, maximum external rotation of the shoulder during the late cocking phase; BR, ball release.

Values indicate Spearman's correlation coefficient (r_s).

Because the KJOC score data were not normal, statistical analysis was performed using Spearman's rank correlation coefficient. a: $p < 0.05$

QB's throwing motion and KJOC score

Table VI. Correlation between trunk, pelvic, and hip–shoulder separation angle during 30-yard throws and KJOC scores

SFC, stride foot contact during the early cocking phase; MER, maximum external rotation of the shoulder during the late cocking phase; BR, ball release

Values indicate Spearman's correlation coefficient (r_s).

Because the KJOC score data were not normal, statistical analysis was performed using Spearman's rank correlation coefficient.

a: $p < 0.05$; b: $p < 0.01$

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Table I. Kerlan-Jobe Orthopedic Clinic (KJOC) score

Item	Question
1	How complicated is it for you to get loose or warm up before a competition or training?
	0: Never feel loose during games or training, 10.0: Normal warm-up time
2	How much pain do you usually experience in your shoulder or elbow?
	0: Pain at rest, 10.0: No pain during competition
3	How much weakness and/or fatigue (i.e., loss of strength) do you feel in your shoulder or elbow?
	0: Weakness or fatigue preventing any competition, 10.0: No weakness; only normal fatigue from competition
4	How unstable does your shoulder or elbow feel during competition?
	0: Routinely popping out, 10.0: No instability
5	How much have arm problems affected your relationship with coaches, management, or agents?
	0: Left team, traded or waived, lost contract or scholarship, 10.0: Not affected at all
6	How much has your arm been affected your throwing motion?
	0: Completely changed; no longer performs the motion, 10.0: No change in motion
7	How much has your velocity and/or power been affected by your arm?
	0: Lost all power; became a finesse or short-distance athlete, 10.0: No change in velocity or power
8	What restrictions do you have in endurance during competition because of your arm?

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	0: Significant limitation, 10.0: No limitation in endurance
9	How much has your throwing control been affected by your arm?
	0: Unpredictability in all throwing, 10: No loss of control
10	To what extent do you feel your arm has impacted your current level of competition in your sport?
	0: Cannot compete; had to switch sports, 10: Competing at desired level

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Table II. Demographic data of the participants

	Mean ± SD	Range
Age (years)	25.1 ± 6.3	18–36
Body mass (kg)	81.9 ± 9.3	67.0–96.0
Height (cm)	179.1 ± 5.3	172.0–186.0
Experience (years)	10.3 ± 5.7	3.0–20.0

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Table III. Trunk, pelvic, and hip-shoulder separation angle during throwing at 20 and 30 yards

	Variable (°)	Throwing phase		
		SFC	MER	BR
20 yards	Trunk nonthrowing side rotation (+)/throwing side rotation (−)	14.6 (18.0)	59.6 (5.2)	89.1 (5.7)
	Pelvic nonthrowing side rotation (+)/throwing side rotation (−)	26.8 (26.0)	67.3 (9.7)	85.3 (6.4)
	Hip–shoulder separation	12.3 (7.7)	4.3 (4.0)	−0.7 (1.2)
30 yards	Trunk nonthrowing side rotation (+)/throwing side rotation (−)	12.1 (31.3)	61.3 (2.7)	85.3 (4.1)
	Pelvic nonthrowing side rotation (+)/throwing side rotation (−)	29.7 (40.5)	67.4 (8.1)	87.3 (6.7)
	Hip-shoulder separation	11.5 (11.0)	7.3 (7.1)	0.1 (2.7)

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Table VI. Correlation between trunk, pelvic, and hip-shoulder separation angles during 30-yard throws and KJOC scores

		Trunk rotation			Pelvic rotation			Hip-shoulder separation		
		SFC	MER	BR	SFC	MER	BR	SFC	MER	BR
KJOC score	Item 1	−0.229	−0.114	−0.005	−0.164	−0.070	−0.100	0.174	0.169	0.070
	Item 2	−0.040	−0.114	0.144	0.025	0.079	0.065	0.342	0.233	0.164
	Item 3	0.145	0.354	0.056	0.359	0.494	0.140	0.662[a]	0.224	0.219
	Item 4	0.636	−0.196	0.516	0.550	0.406	0.760[b]	0.411	0.507	0.779[b]
	Item 5	0.335	−0.139	0.150	0.335	0.173	0.439	0.26	0.173	0.752[b]
	Item 6	−0.200	0.029	0.162	−0.257	0.010	0.153	−0.124	0.105	0.315
	Item 7	0.563	−0.095	0.687[a]	0.429	0.439	0.839[a]	0.296	0.515	0.734[a]
	Item 8	0.287	0.234	0.488	0.239	0.445	0.478	0.382	0.292	0.368
	Item 9	0.249	0.253	0.564	0.153	0.416	0.535	0.249	0.282	0.368
	Item 10	0.388	0.263	0.194	0.462	0.517	0.249	0.686[a]	0.328	0.263
	Total	−0.210	0.105	0.265	−0.187	0.128	0.260	0.078	0.151	0.425

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Table V. Correlation between trunk, pelvic, and hip-shoulder separation angles during 20-yard throws and KJOC scores

		Trunk rotation			Pelvic rotation			Hip–shoulder separation		
		SFC	MER	BR	SFC	MER	BR	SFC	MER	BR
KJOC score	Item 1	−0.333	0.244	−0.154	−0.194	0.219	−0.124	0.263	0.144	0.252
	Item 2	−0.178	0.392	0.010	−0.060	0.377	0.030	0.332	0.233	0.311
	Item 3	−0.019	0.709[a]	−0.047	0.271	0.592	0.000	0.564	0.378	0.311
	Item 4	0.540	0.129	0.454	0.277	0.292	0.511	0.053	0.373	0.172
	Item 5	0.231	−0.041	0.035	0.081	−0.052	0.162	−0.121	0.185	0.249
	Item 6	−0.343	0.181	0.114	−0.420	0.305	0.172	−0.038	−0.162	0.253
	Item 7	0.448	0.229	0.601	0.153	0.343	0.658[a]	−0.029	0.267	0.115
	Item 8	0.081	0.535	0.378	−0.024	0.574	0.406	0.234	0.115	0.273
	Item 9	0.062	0.488	0.454	−0.110	0.507	0.483	0.110	0.019	0.158
	Item 10	0.179	0.522	0.070	0.254	0.616[a]	0.099	0.562	0.388	0.503
	Total	−0.393	0.475	0.137	−0.370	0.397	0.219	0.050	−0.064	0.201

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Table IV. Total KJOC score and individual item score

Variables	Score
Item 1	10.0 (0.6)
Item 2	10.0 (1.0)
Item 3	9.8 (1.7)
Item 4	9.9 (0.6)
Item 5	10.0 (0.2)
Item 6	9.7 (3.1)
Item 7	9.9 (0.5)
Item 8	9.8 (1.4)
Item 9	9.8 (2.5)
Item 10	10.0 (0.5)
Total	92.7 (8.7)

Journal Pre-proof

Journal Pre-proof